Image Classification Extension

https://github.com/aiyshwary/Image-Classification-Extension-in-Rapid-Miner

OVERVIEW

A RapidMiner Studio extension enabling no-code transfer learning for image classification, supporting fin
on custom datasets and ONNX-based inference without writing a single line of code.

IMPACT

Developed a RapidMiner extension with two custom operators — Fine Tuning and Inference — enabling
with ResNet, VGGNet, DenseNet, Inception and more using a drag-and-drop interface.

KEY DETAILS

1. Fine Tuning operator accepts an image directory and hyperparameters; outputs a fine-tuned model in
2. Inference operator loads the ONNX model and outputs a prediction DataFrame with classified labels.
3. Supports ResNet, AlexNet, VGGNet, SqueezeNet, DenseNet, and Inception architectures.
4. Compatible with both CPU and GPU (CUDA) environments via configurable conda environment.
5. Installed as a JAR extension in RapidMiner Studio with Python Scripting backend.
6. Inference operator requires four inputs: the ONNX model path, image path, class-label text file, and res
output images alongside the prediction DataFrame.
7. Depends on Custom Operators (>=1.1.0) and Python Scripting (>=10.0.1) marketplace extensions wit
3.10.8, ONNX 1.15.0, ONNX Runtime 1.15.1).